

Kenny Nguyen

US Citizen - Gilbert, Arizona

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Education

- BS** **Arizona State University**, BS in Computer Science Aug 2022 – May 2026
- GPA: 3.88/4.0
 - Relevant Coursework: Foundations of Algorithms, Operating Systems, Computer Networks, Data Structures and Algorithms, Software QA and Testing, Principles of Programming Languages

Experience

- Tokyo Electron America**, Software Engineering Intern (Contract) Aug 2025 – May 2026
- Designed and implemented a C# state machine in Unity governing NPC behavior with multiple states and transition conditions; wired event-driven communication between subsystems to trigger cross-component behavior on state changes.
 - Led final integration of independently developed Unity modules into a single Meta Quest VR application using the Meta XR SDK, establishing correct state initialization, scene transitions, and end conditions for Android/Quest standalone deployment.
 - Debugged cross-module integration failures in C#/Unity where components passed in isolation but broke in combined runtime; resolved Quest 2 and Quest 3 device-specific geometry collision failures and UI layout inconsistencies.
- Opportunity Hack**, Software Engineer Intern – Remote June 2025 – Aug 2025
- Led a 3-person team to design and ship a full-stack alumni platform serving 1,000+ users, making key architectural decisions including technology selection and system design.
 - Designed the platform's data model and access control architecture, choosing Row Level Security (RLS) for role separation and Clerk for authentication with webhook-based account linking.
 - Delivered an interactive alumni world map with real-time filtering, a CSV bulk import tool that migrated 1,000+ legacy records, and an admin dashboard for database management and bulk email campaigns.

Projects

- Distributed Order Processing System | Java, Spring Boot, Kafka, PostgreSQL, Docker**
- Built an event-driven microservices system with 3 Spring Boot services (order, stock, email) communicating asynchronously via Kafka, persisting orders to PostgreSQL and fanning out to downstream consumers via a shared domain event model.
 - Containerized all services with Docker Compose including health checks and service dependency ordering, enabling one-command local deployment of the full Kafka + PostgreSQL + microservices stack.
- Distributed Storage System | Python, UDP Sockets**
- Designed a 3-process distributed storage system in Python (manager coordinator, user client, and multi-threaded disk nodes) communicating over UDP sockets via a custom 23-command protocol.
 - Implemented XOR-based parity with rotating parity blocks across N disks, enabling automatic single-disk failure recovery by reconstructing lost data from N-1 surviving blocks using parallel threaded I/O.
- Redis Server Implementation | Go, TCP Sockets**
- Implemented a Redis-compatible server in Go supporting 9 commands including key-value storage with TTL expiration and Redis Streams with blocking reads, range queries, and auto-generated entry IDs by parsing the RESP protocol over raw TCP.
 - Handled concurrent client connections using a goroutine-per-client model with mutex-protected shared storage, serving multiple simultaneous clients on port 6379 with no external dependencies.

Skills

Languages: Python, Go, Java, C/C++, JavaScript, TypeScript, HTML/CSS, Bash, SQL

Frameworks & Databases: Spring Boot, React, JUnit, Mockito, PostgreSQL

Tools & Concepts: Kafka, Docker, Git, Linux/Unix, Cursor, Claude Code, REST APIs, Microservices